

Project Proposal

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I will be working alone for this project. The dataset I will be using comes from scorenet-work.org. This is a website that displays a variety of statistics for different divisions, levels and leagues for a large amount of sports in the USA.

For this project, I will be using the team statistics for NCAA D1 Women's Lacrosse. The data includes 120 different division one teams and their statistics from the 2022-2023 season. With this data, I plan to group the teams geographically by region/conference. Then, I want to look at some trends among and within each conference.

```
library(tidyverse)
```

```
lax_women <- read_csv("/Users/hadleystabnau/Desktop/STAT408/data/lacrosse_women_ncaa_div1_2022-2023.csv")
```

For this intro, I will just plot the data looking at each team's average goals per game vs their win % as a team.

```
names(lax_women)
```

```
[1] "Team" "draw_pctg"
[3] "turnovers_per_game" "sog_per_game"
[5] "assists_per_game" "caused_turnovers_per_game"
[7] "clearing_pctg" "draw_controls_per_game"
[9] "fouls_per_game" "free_position_pctg"
[11] "ground_balls_per_game" "points_per_game"
[13] "save_pctg" "goals_allowed_per_game"
[15] "goals_per_game" "shots_per_game"
[17] "shot_pctg" "win_pctg"
```

```
lax_women |>
  ggplot(aes(x=goals_per_game, y=win_pctg)) +
  geom_point() +
  labs(title= "Womens D1 Lacrosse", x="Average Goals Per Game", y="Win %")
```

